

Corentin Dupaigne

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Education

Epita

Sep 2024 — Aug 2027

Master of Engineering (M.Eng.) in Computer Science

Paris

• Key Academic Projects:

- POSIX-compliant Shell (Lexer/Parser, AST, Expansion engine, Process execution & redirections)
- Full-Stack microservice app deployed with Kubernetes, CICD and Ansible.

• Relevant Coursework: Operating Systems, DSA, Cloud & DevOps, Networking, Algorithmic Complexity, Data Compression, Graph Theory, Object-Oriented Programming (C++/Java), SQL, REST, Design Patterns, Layered Architecture, Agile.

Work Experience

Full-Stack & DevOps Engineer (Apprentice)

September 2024 — Present

Enedis

Paris

- Streamlined maintenance for 14 internal applications, managed and optimized databases of thousands of users.
- Reduced technical debt by migrating legacy PHP tools to Laravel and Next.js, Containerized legacy apps to ease future development.
- Reduced deployment toil and environment drift by implementing Ansible and GitLab CI pipelines.
- Designed and deployed a modern web application with decoupled backend (Laravel) and frontend (React) architecture
- Reduced vulnerability count from 4000+ to 20 across 14 applications by implementing Checkmarx SAST and DAST pipelines integrated into GitLab CI

Projects

Condensation

- Designed and implemented multi-environment Kubernetes infrastructure (dev/prod) using Kustomize overlays on a self-managed VPS (OVH)
- Implemented reusable GitHub Actions workflows for CI/CD: automated lint, tests and vulnerability scanning
- Automated CD pipeline: Docker images built and pushed to GHCR on merge, deployed to k3s via SSH with automated rolling updates
- Configured cluster networking with Gateway API (Envoy-based) for traffic routing across microservices
- Built OAuth2/PKCE authentication microservice (Laravel)
- Architected event-driven microservices platform with Kafka as event bus

Homelab

- Fully reproducible bare-metal k3s cluster provisioned with Ansible in a single command
- Hosting 4 production applications across multiple domains with automated rolling deployments on every push via GitHub Actions and GitLab CI
- Traffic routing managed with Gateway API (Envoy-based)
- Exposed Gateway API (Envoy) to external traffic using MetalLB as bare-metal load balancer
- Observability stack with Prometheus and Grafana monitoring cluster metrics

Skills

- **Programming Languages:** Go, C/C++, Python (Fast API, Django), Bash, JavaScript, PHP (Laravel), SQL.
- **Cloud & Orchestration:** Kubernetes (k3s, Kustomize, Helm, Gateway API), Docker, AWS.
- **IaC:** Ansible.
- **CI/CD:** GitLab CI, GitHub Actions.
- **Networking:** TCP/IP, DNS, HTTP/TLS, Nginx.
- **Linux:** Debian/Fedora, vim.